GENERAL RESEARCH

Estimation of Total Entropy of Melting of Organic Compounds

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The aim of this study is to provide a simple means of estimating the total entropy of melting (ΔS_m^{tot}) for a wide range of pharmaceutically and environmentally relevant organic compounds. A semiempirical equation based on only 2 molecular descriptors, the rotational symmetry number(σ) and the molecular flexibility number (Φ), has been used to calculate ΔS_m^{tot} for 1799 organic compounds. The average absolute error in estimating ΔS_m^{tot} is 12.3 J/K·mol. This method gives entropy predictions that are comparable to those of a recently published group additivity method that utilizes 144 group contribution values.

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Introduction

The estimation of the physicochemical properties of organic compounds can provide useful insight into their pharmaceutical efficacy and environmental exposure limits. The entropy of melting is an important property in predicting the melting point and solubility of a compound.[1,2] The total entropy of melting (ΔS_m^{tot}) includes entropies of all solid−solid transitions and the solid−liquid transition. According to Bondi[3], the total entropy of melting of a compound is the sum of its positional, rotational, and conformational components:

$$\Delta S_m^{tot} = \Delta S_m^{pos} + \Delta S_m^{rot} + \Delta S_m^{conf} \qquad \text{(I)}$$

A number of approaches have been used for estimating the total entropy of melting of organic compounds. According to Walden's rule, the entropy of melting for aromatic compounds with little flexibility (e.g., diphenyl and cyclopentane) is constant with a value of 56.5 J/K·mol. Richard's rule states that the entropy of melting is constant for small spherical compounds (e.g., methane and neon) with a value of 10.5 J/K·mol. A large number of compounds do not belong to either of the above categories, and, hence, cannot be reasonably estimated.[4] Recently, Chickos et al. developed a group additivity method[5] for estimating the total entropy of melting. Although it has wide applicability, the method is cumbersome as it employs a large number (144) of group contribution values and values may be missing for some group fragments.

Dannenfalser and Yalkowsky[4] developed the following semiempirical equation based on only two nonadditive molecular descriptors, the rotational symmetry number (σ) and the flexibility number (Φ), to estimate the total entropy of melting for nonelectrolytes:

$$\Delta S_m^{tot} = C - R\ln\sigma + R\ln\Phi \text{ J/K·mol} \qquad \text{(II)}$$

In this paper, the above equation is applied to a complex set of data containing a wide range of pharmaceutically and environmentally relevant organic compounds. The total entropy of melting (ΔS_m^{tot}) values obtained using the above semiempirical equation have been compared to those calculated by the group additivity method of Chickos et al.[5]

Methods

Molecular Rotational Symmetry Number. Rotational symmetry number (σ) is defined as the number of positions into which a molecule can be rotated that are identical to a reference position. Examples of rotational symmetry for a number of molecules are shown in Figure 1.

While assigning a value to σ, groups such as methyl, hydroxyl, mercapto, and primary amines are assumed to be freely rotating and are treated as being radially symmetrical. The nitro and carboxyl groups are treated as laterally symmetrical. All other groups are assumed to be asymmetrical. The rotational symmetry number for a molecule is never less than unity because every molecule has at least one identical orientation that is produced by a 360° rotation about any axis. Rings with six or fewer atoms are considered to be rigid and symmetrical. For example, cyclopropane, cyclobutane, and cyclopentane are assigned symmetry numbers of 6, 8, and 10, respectively.

Conical (e.g., hydrogen cyanide and chloromethane) and cylindrical (e.g., carbon dioxide and ethane) molecules are empirically assigned symmetry numbers of 10 and 20, respectively. Spherical molecules (e.g., neon and methane) are assigned a symmetry number of 100. All flexible molecules are assigned a symmetry number of unity.[4]

Molecular Flexibility Number. For a simple linear molecule, the flexibility number (Φ) is an exponential function of its chain length and is defined as

$$\Phi = 2.435^{\tau} \qquad \text{(III)}$$

where τ is the number of torsional angles or the

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10.1021/ie0497745 CCC: $27.50 © 2004 American Chemical Society
Published on Web 05/28/2004

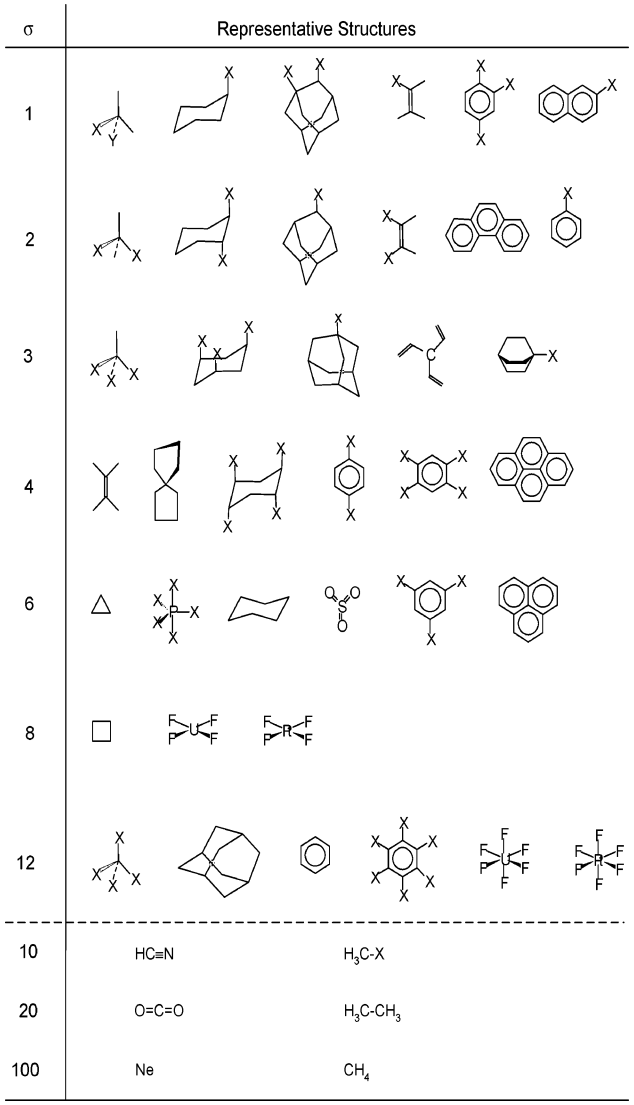


Figure 1. Symmetry numbers for several molecules.

flexibility count. The flexibility count, τ, for any compound can be calculated using the following semiempirical equation of Dannenfelser and Yalkowsky:[6]

$$\tau = SP^3 + 0.5SP^2 + 0.5Ring - 1 \qquad \text{(IV)}$$

where $SP^3 = \sum$ Nonring(CH_2,CH,C,NH,N,O,S), $SP^2 = \sum$ Nonring(=CH,=C,=N,C=O), and Ring = $\sum$ independent single, fused, or conjugated ring systems.

Substituting eq IV into eq III we get

$$\Phi = 2.435^{[SP^3+0.5SP^2+0.5Ring-1]} \qquad \text{(V)}$$

The value 2.435 is based upon theoretical calculations by Tonelli,[7] Kirshenbaum,[8] and Starkweather and Boyd,[9] which state that the conformational entropy of melting for an isolated polyethylene chain is 7.4 ± 0.2 J/K·mol per mole of monomer. These calculations are based on the assumptions that the trans conformation is more stable than the gauche conformations and the gauche (+) − gauche (−) sequences are hindered or forbidden. Experimental data reported in refs 7−9 give similar values.

Terminal groups, such as CH_3, NH_2, OH, CN, F, Cl, Br, I, =O, $=CH_2$, and ≡N, as well as nonterminal sp hybrid atoms do not contribute to molecular flexibility and are therefore not included in the calculation of Φ. Also, carbons with three identical groups, for example, trihalomethyl or *tert*-butyl, are not counted while determining Φ. Compounds with a negative value for τ are assigned a value of 0. Aliphatic cyclic compounds with fewer than eight carbon atoms are counted as a single ring system. A few examples of molecular flexibility numbers are shown in Table 1.

Table 1. Examples of Some Molecular Flexibility Numbers

name	σ	SP^3	SP^2	ring	τ	Φ
n-pentane	1	3	0	0	2	5.93
2-methylbutane	1	2	0	0	1	2.44
neopentane	1	1	0	0	0	1.00
1-pentanol	1	4	0	0	3	14.44
3-pentanone	1	2	1	0	1.5	3.80
cyclopentane*	10	0	0	1	0	1.00
toluene	2	0	0	1	0	1.00
tert-butylbenzene	2	0	0	1	0	1.00
1,2,4 trimethylbenzene	1	0	0	1	0	1.00
1,3,5-trimethylbenzene	6	0	0	1	0	1.00
4-chlorophenol	2	0	0	1	0	1.00
benzoic acid	2	0	0	1	0	1.00
2,4,6-trinitrotoluene	2	0	0	1	0	1.00
n-butylbenzene	1	3	0	1	2.5	9.25
hexamethylbenzene	12	0	0	1	0	1.00

Data

Experimental total entropies of melting ($\Delta S^{tot}_{m,exp}$) and total entropies of melting predicted by their group additivity method ($\Delta S^{tot}_{m,ga}$) for 1984 compounds were compiled by Chickos et al.[4] and entered into an MS-EXCEL worksheet. Of these, 1799 compounds with $\tau \leq 18$ were selected for generating the statistics. Compounds with $\tau > 18$ were not included because of large variations observed in the data for these very long chain compounds. Aliphatic cyclic compounds with eight or more carbons in the ring were not included in the calculations. Additionally, compounds with unreasonably low reported entropy values (less than 8.0 J/K·mol) were excluded from our dataset. Molecular rotational symmetry and flexibility numbers were assigned to each compound as described above.

Results and Discussion

The total entropy of melting (ΔS^{tot}_m) was calculated using the following equation developed by Dannenfelser and Yalkowsky:[6]

$$\Delta S^{tot}_m = 50 - R\ln\sigma + R\ln\phi \text{ J/K·mol} \qquad \text{(VI)}$$

The total entropy of melting data for these 1799 compounds are listed in the Supporting Information.

The average absolute errors in predicting the total entropy of melting for these compounds using eq VI and the group additivity method are 12.3 and 10.4 J/K·mol, respectively. The error distribution obtained while correlating the experimental reported values ($\Delta S^{tot}_{m,exp}$) and entropy values predicted by eq VI (ΔS^{tot}_m) is shown in Figure 2.

As can be seen in Figure 2, the error distribution is fairly symmetrical with a few outliers. A majority of the outliers are the very long chain flexible molecules. Such compounds are very likely to form liquid crystals and exhibit mesophasic behavior and multiple transitions before the actual melting to the isotropic liquid. Some-

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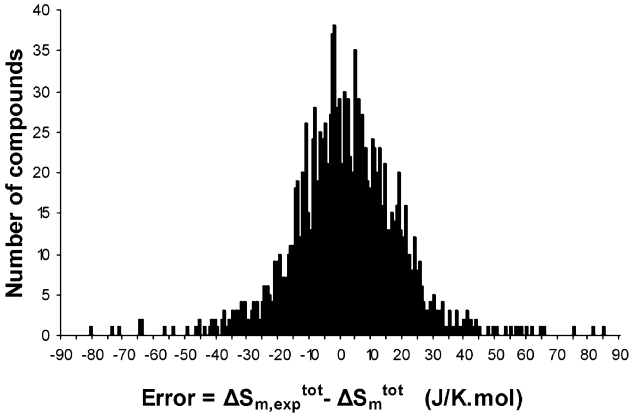


Figure 2. Distribution of errors in predicting entropy of melting.

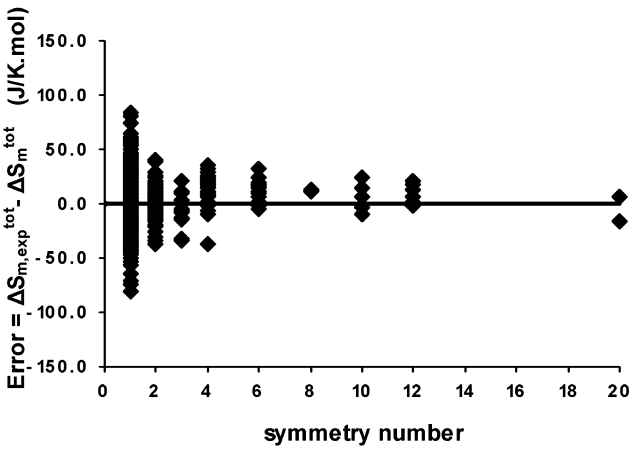


Figure 3. Error distribution as a function of symmetry number (σ).

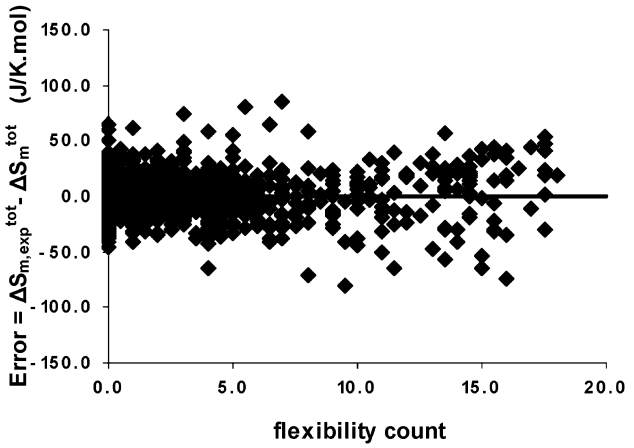


Figure 4. Error distribution as a function of flexibility count (τ).

times the entropies associated with all these transitions are not accounted for in the reported experimental entropy values. This leads to an increase in the average error associated with their entropy predictions.

Figures 3 and 4 show the distribution of errors as a function of rotational symmetry number (σ) and molecular flexibility count (τ) for the entire set of compounds. In Figure 3, an almost symmetric distribution of errors is observed at each rotational symmetry number. On the other hand, as shown in Figure 4, at low flexibility counts the errors are distributed symmetrically, but as the flexibility count increases the distribution becomes more random with a large proportion of outliers. This suggests that because of their tendency to exhibit mesophasic and liquid crystalline behavior, the entropy of very long flexible compounds cannot be accurately predicted using the proposed equation.

Table 2. Average Absolute Errors in Entropy Prediction

τ	no. of compounds	average absolute error (J/K·mol)	
		equation 7	group additivity
≤12	1722	11.7	9.9
≤18	1799	12.3	10.4

To further verify the effect of long flexible molecules on the errors in entropy prediction, 77 compounds with τ between 12 and 18 were excluded from the calculations. As expected, the absolute average errors for our semiempirical equation and group additivity method decreased to 11.7 and 9.9 J/K·mol, respectively, when these long flexible molecules were not included in the calculations. The results are shown in Table 2.

Despite the inability to take into account the mesophasic and liquid crystalline behavior of a few very long flexible molecules, our proposed equation does well in predicting the entropy of melting for a complex data set of nearly 1800 organic compounds. The use of only two molecular descriptors in our equation is an advantage over the group additivity method which requires 144 group contribution values.

Conclusion

The semiempirical equation of Dannenfelser and Yalkowsky (eq VI) has been validated and applied to a large data set of pharmaceutically and environmentally relevant organic compounds. This equation predicts the total entropies of melting for 1799 organic compounds with an average absolute error of 12.3 J/K·mol. It is a reasonable estimate considering that no coefficients were generated from experimental data for a training set.

Supporting Information Available: A tabulation of the experimental ($\Delta S_{m,exp}^{tot}$) and predicted total entropies of melting (ΔS_{m}^{tot} and $\Delta S_{m,ga}^{tot}$) for 1799 compounds used in the study along with the rotational symmetry number (σ) and flexibility count (τ). This material is available free of charge via the Internet at http://pubs.acs.org.

Nomenclature

ΔS_m^{tot} = total entropy of melting predicted using eq VI, J/K·mol
ΔS_m^{pos} = positional entropy of melting, J/K·mol
ΔS_m^{rot} = rotational entropy of melting, J/K·mol
ΔS_m^{conf} = conformational entropy of melting, J/K·mol
$\Delta S_{m,exp}^{tot}$ = experimental total entropy of melting, J/K·mol
$\Delta S_{m,ga}^{tot}$ = total entropy of melting predicted by group additivity method, J/K·mol
R = universal gas constant, 8.314 J/K·mol

Greek Symbols

σ = molecular rotational symmetry number
Φ = molecular flexibility number
τ = flexibility count or number of effective torsional angles

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Received for review March 22, 2004
Revised manuscript received April 28, 2004
Accepted April 30, 2004

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